

Physics A (H156, H556)

Diffraction grating ppq

KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname								
Centre number						Candidate number				

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **56**.
- The total number of marks may take into account some 'either/or' question choices.

- 1 Monochromatic light of wavelength 600 nm is passed through a diffraction grating.

Maxima are formed on a screen 3.0 m from the grating. The distance between the two first order maxima, on either side of the zero maximum, is 1.42 m.

What is the spacing, in m, between lines on the diffraction grating?

- A 1.3×10^{-6}
- B 1.4×10^{-6}
- C 2.5×10^{-6}
- D 2.6×10^{-6}

Your answer

[1]

- 3 Light from a hydrogen source is incident normally at a diffraction grating. The first order maximum of the H-alpha spectral line of wavelength 486 nm is observed at angle of 30.0° .

Light from a distant receding star is observed using the same diffraction grating. The light is incident normally at the grating as before. The speed of this star is $0.16c$, where c is the speed of light in a vacuum.

What is the observed angle of the first order maximum of the H-alpha spectral line from the light of this receding star?

- A 24.8°
- B 30.0°
- C 34.8°
- D 35.5°

Your answer

[1]

- 4 Laser light of wavelength of 640 nm is incident normally at a diffraction grating. The separation between adjacent lines (slits) is 3.3×10^{-6} m.

What is the **total** number of bright spots that can be observed in the diffraction pattern?

- A 5
- B 6
- C 10
- D 11

Your answer

[1]

5 Hydrogen atoms excited in a discharge tube only emit four different discrete wavelengths of visible photons.

*In a semi-darkened room, a single slit is placed in front of the discharge tube. A student holds a diffraction grating which has 300 lines per millimetre.

The student looks through the grating at a 15 cm plastic ruler placed 0.50 m away, as shown in Fig. 5.1.

The paths of the different colours of light from the slit to the student's eye are shown in Fig. 5.2.

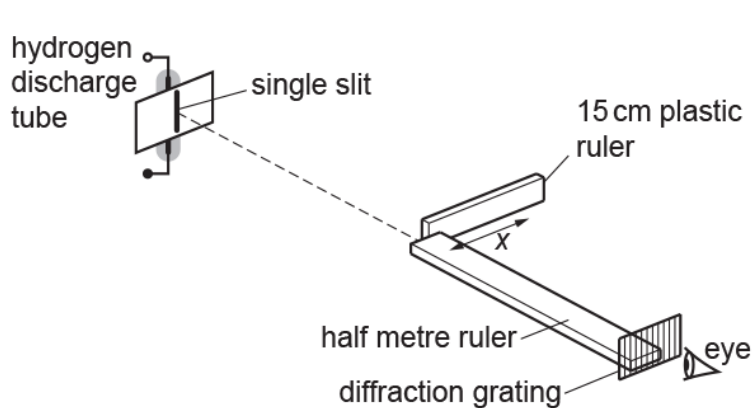


Fig. 5.1 (not to scale)

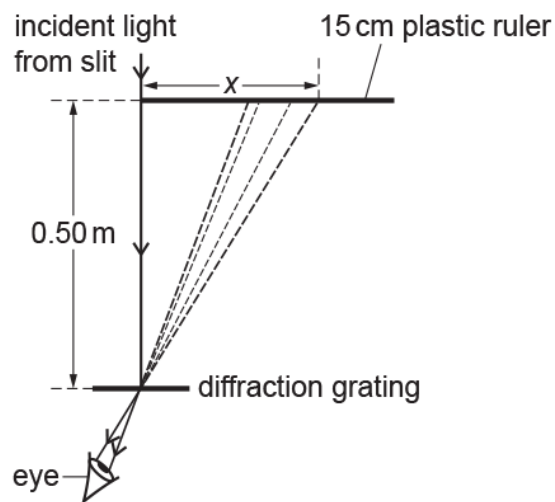


Fig. 5.2 (not to scale)

Four **first** order images of the slit, one at each photon wavelength, are observed as vertical lines against the background of the plastic ruler, as shown in Fig. 5.3.

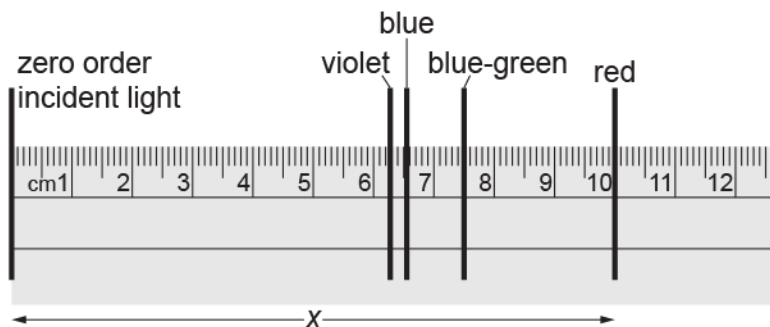


Fig. 5.3

The student decides to determine the wavelength of the photons which form the red line observed at $x = 10$ cm on the ruler.

- Describe how the information that has been given can be used to determine the wavelength of the red photons.
- Estimate the percentage uncertainty in the measured value of the wavelength.

This image shows a full-page view of a document template. It consists of a white background with approximately 20 horizontal dashed lines spaced evenly apart, typical of primary or secondary school writing paper. In the bottom right corner, there is a small black rectangular box containing the white text "[6]", indicating the page number.

Fig. 24.1 shows the laser beam incident normally at a diffraction grating.

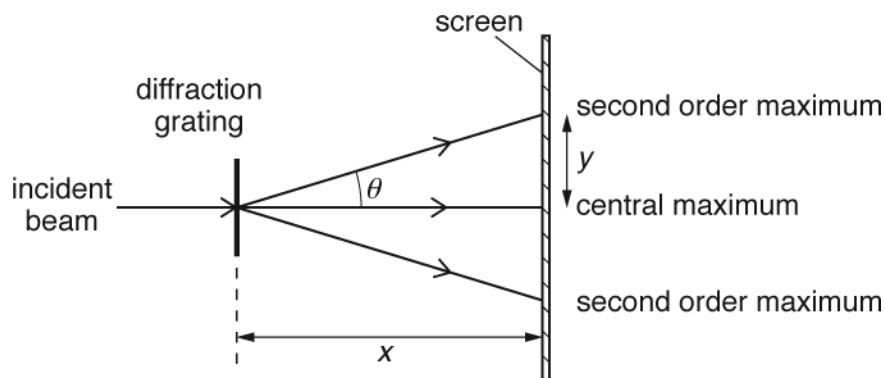


Fig. 24.1

The students use a diffraction grating with $600 \text{ lines mm}^{-1}$. They vary the distance x between the grating and the screen from 1.000 m to 2.000 m . They measure the distance y from the *central* maximum to the *second order* maximum.

The students decide to plot a graph of y against $\sqrt{x^2 + y^2}$.

Show that the gradient of the graph is equal to $\sin \theta$, where θ is the angle between the central maximum and the *second order* maximum.

[1]

(b) Fig. 24.2 shows the graph plotted by the students.

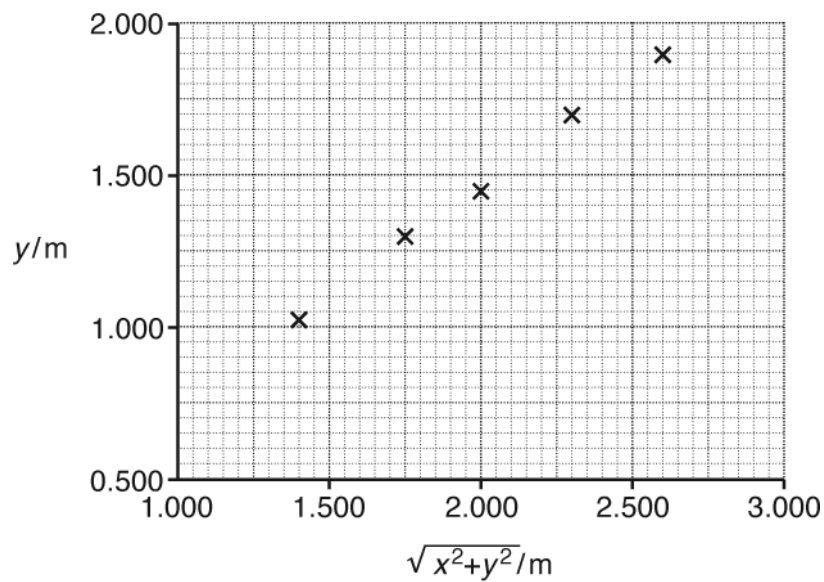


Fig. 24.2

- i. Use Fig. 24.2 to determine an accurate value of the wavelength λ of the light from the laser.

$\lambda =$ _____ m [2]

- ii. Suggest why there are no error bars shown in Fig. 24.2.

 _____ [1]

- iii. Suggest how the precision of this experiment may be affected by using a protractor to measure the angle θ .

 _____ [1]

7(a)

Light from a distant galaxy is passed through a diffraction grating. Fig. 21.2 shows the part of the spectrum of light that shows a strong hydrogen-alpha emission line.

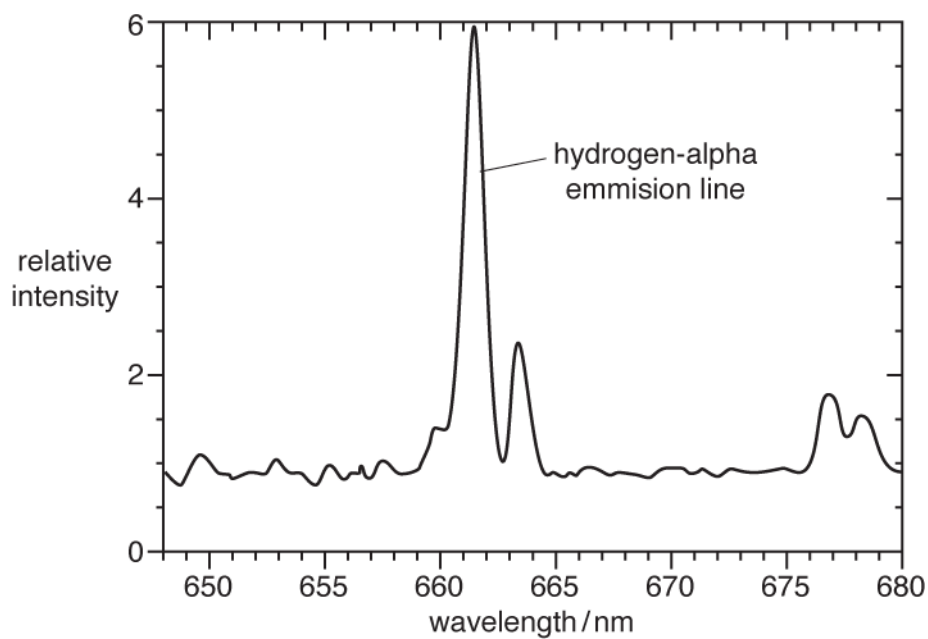


Fig. 21.2

- i. State how an emission line is produced.

[1]

- ii. State an adjustment that could be made to the experimental arrangement that would space the emission lines more widely.

[1]

- iii. In the laboratory, the wavelength of the hydrogen-alpha emission line is 656.3 nm. Use Fig. 21.2 to determine the recession velocity of the galaxy.

recession velocity = _____ m s⁻¹ [3]

- iv. Suggest why hydrogen spectral lines play an important role in determining red shift of galaxies.

----- [1]

8 Algol is a triple-star system, with stars Aa1, Aa2 and Aa3 orbiting each other. This triple-star is 90 light-years from the Earth.

The table shows some data about the three stars of Algol.

Star	Luminosity of star / $L_{\odot}$	Surface temperature of star / K
Aa1	182	13 000
Aa2	6.92	4500
Aa3	10.0	7500

The luminosity of each star is in terms of the solar luminosity $L_{\odot}$.

i. Define the **luminosity** of a star.

[1]

ii. Use Stefan’s law to determine the ratio $\frac{\text{radius of star Aa2}}{\text{radius of star Aa3}}$.

ratio = [2]

iii. Use Wien’s displacement law to explain which star would have the **longest** wavelength at the peak intensity of the emitted electromagnetic radiation.

[2]

iv. Suggest how an astronomer using just an optical telescope can deduce that the three stars of Algol have different surface temperatures.

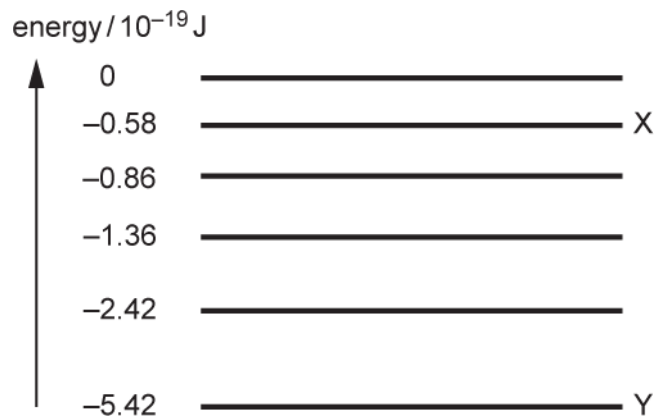
[1]

-
-
-
-
-

[3]

- 9 The diagram shows some of the energy levels of the electron in a hydrogen atom.

energy / 10^{-19} J



The light from the stars in a distant galaxy is analysed on the Earth using a diffraction grating. Dark lines are observed in the spectrum.

An astronomer concludes that the dark line at a wavelength 432 nm corresponds to the electron transition between X and Y.

- i. Explain the origin of the dark lines.

----- [2]

- ii. Calculate the recession velocity v of the galaxy.

$$v = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- iii. State the name of the theory that is supported by evidence from the measurement of the recession velocities of galaxies in the universe.

----- [1]

10 This question is about the brightest wavelength (590 nm) of light from a sodium lamp.

Analysis of the light from the sodium lamp using a diffraction grating shows that there are photons of two different energies at wavelengths 589.0 nm and 589.6 nm.

- i. Calculate the energy difference ΔE between these two photons.

$$\Delta E = \text{-----} \text{ J [3]}$$

- ii. The light at these wavelengths can be seen as two separate lines when viewed through a diffraction grating. In order to be distinguishable from each other, the angular separation between the lines must be at least 0.02° .

Show that the lines will appear separated in the **second order** spectrum when the sodium lamp is viewed through a grating with 300 lines per millimetre.

[3]

- 11 The Big Bang theory explains the origin and the evolution of the early Universe.

The table below shows the distance d and recession velocity v of some galaxies close to our own galaxy.

Galaxy	d / Mpc	$v / \text{km s}^{-1}$
NGC-5357	0.45	200
NGC-3627	0.90	650
NGC-4151	1.7	960
NGC-4472	2.0	850

The chemical composition of the stars in our galaxy can be determined by analysing in the laboratory the absorption spectral lines for these stars.

The closest star to us is the Sun. The wavelength of the hydrogen-beta spectral line from the Sun is 486 nm.

- i. Use the information from the table to calculate the **observed** wavelength λ of the hydrogen-beta spectral line from a star in the galaxy NGC-4151.

$$\lambda = \text{----- nm [3]}$$

- ii. A diffraction grating with 800 lines per mm is used to observe and analyse the light from the Sun in the laboratory.

A narrow beam of light from the Sun is incident normally at the diffraction grating.

Calculate the angle θ between the central beam of light through the grating and the hydrogen-beta spectral line in the **second** order spectrum.

$$\theta = \text{----- } ^\circ \text{ [2]}$$

12 When the light from a star is passed through a diffraction grating it forms a spectrum.

Which of the following statements is / are correct?

1. Light emitted from the surface of a star would form a continuous spectrum.
2. Light received from the Sun has dark lines across its spectrum which correspond to the absorption of certain wavelengths by atoms in the Earth's atmosphere.
3. A photon in an emission spectrum occurs when an electron moves from a low to a higher energy level within an atom.

- A. 1, 2 and 3
B. Only 1 and 2
C. Only 2 and 3
D. Only 1

Your answer

[1]

END OF QUESTION PAPER

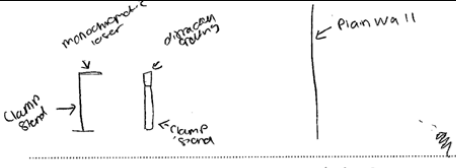
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			D	1	Examiner's Comments Candidates often confused which formula to use. Some used the two-slit formula and others, correctly, used the diffraction grating formula. Here using $d \sin q = n\lambda$ is the correct approach. The question gives the distance between the two first-order maxima. This means that the distance from the centre to the first-order maximum is <i>half</i> of that i.e. 0.71 m. Consequently, the angle q is $\tan^{-1}(0.71 \div 3.0) = 13.3\dots$ degrees. Substituting in the wavelength (600×10^{-9} m) and $n=1$ gives d as 2.61×10^{-6} m. Therefore the answer is D.
			Total	1	
2			Level 3 (5–6 marks) description and analysis, with an appropriate comment about accuracy <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Description and some graphical analysis and some comment about accuracy <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Incomplete description or analysis, or an appropriate comment about accuracy <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark <i>No response or no response worthy of credit.</i>	B1 × 6	NB - use of two slit interference formula is incorrect physics. Mark XP and treat description as incomplete Indicative scientific points may include: Description • Angles or distances measured from central maximum • Some method of measuring angle directly, e.g. protractor • Measuring distances from grating to screen and from central maximum to others • Description of maxima • White screen or wall • Recognition and mitigation of risk of injury to eye Analysis • Graph of d sin theta against n with gradient lambda (or workable alternative) • Conversion of lines per mm to distance between lines Accuracy • Use large room to achieve large distances • Large distance reduces relative uncertainty • Measure distance between corresponding maxima and halve • Use of trigonometry to convert

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					measured distances into angles from the centre • use spots either side of the centre • change diffraction grating for smaller d so that maxima are more spaced out • change diffraction grating for larger d so that there are more maxima (and so more points on the graph) • darken the room <u>Examiner's Comments</u> This experiment is one of the suggested practicals to complete in the Practical Endorsement. The relevant formula was not on the question paper itself, however it is in the relationships and formulae booklet in the appropriate place. Candidates that described the two-slit interference experiment could not, therefore, be awarded any more than Level 1 (2 marks). As explained in previous series, approaches required a graphical element to find the wavelength (amongst other aspects) in order to achieve Level 2 (4 marks). Finally, a key aspect that distinguished between candidates was their ability to describe and explain the necessary precautions for high quality data. A selection of these is listed in the Mark Scheme. Exemplar 1

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			 • set up equipment shown in a dark room • calculate small distance by • measure distance from clamp stand 1 to clamp stand 2 and also vertical distance to screen using ruler • calculate $\sin \theta$ using trig. calc's. number of maxima minima • also determined by $\frac{1}{\lambda} \times 10^{-3}$ number of max given • measure distance between central bright spot and adjacent bright spot (not over) • repeat for different values of $\sin \theta$ • plot graph of also $\sin \theta$ against distance between spots • gradient = $n \lambda$ • improve accuracy by taking measurements multiple times and calculating mean. Exemplar 1 has satisfied all of the criteria for a Level 3 response, 6 marks. The candidate has described the correct experiment and made it clear how they are going to measure the relevant quantities such as the distance of the maxima away from the centre and the distance to the screen so that they can later calculate theta and hence $\sin(\theta)$. This already is promising for a Level 3 response. The candidate has also completed the analysis correctly by using a graphical approach. It is clear what is going to be plotted on which axis and what the gradient of the line of best fit will represent. Finally, it is clear how the wavelength shall be calculated. The candidate has used trigonometry to calculate theta and has stated that the room should be dark, All of these factors taken together indicate Level 3 thinking.
	Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3			D	1	<u>Examiner's Comments</u> Candidates found this question challenging because it joins two ideas together. The first idea is that of the Doppler effect on waves from a moving source, which will increase the wavelength received by the observer. This removes options A and B. Once the new wavelength has been calculated, the corresponding angle should be calculated from the diffraction grating equation. It is worth mentioning that the correct answer and the most likely incorrect answer are too close together for the candidate to merely guess.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4			D	1	
			Total	1	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
5	Level 3 (5 - 6 marks) Clear procedure or correct determination of wavelength, plus reasonable estimation of uncertainty in λ or $(\sin) \theta$ <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3 – 4 marks) Description of procedure or correct determination of λ, but no estimation of uncertainty or Clear estimation of uncertainty in wavelength but limited description of procedure and/or determination of λ or $(\sin) \theta$ or Some description of procedure, an attempt to determine the wavelength, and an attempt to estimate uncertainty in some of the measurements (e.g. in x) <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1 – 2 marks) A limited selection from the scientific points worthy of credit. <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks No response or no response worthy of credit. <u>Frontal</u>	1 (AO3)	Use level of response annotations in RM Assessor, e.g. L2 for 4 marks, L2⁺ for 3 marks, etc. <u>L1 maximum for any answers which use formula $\lambda = ax/D$</u> Indicative scientific points may include: Procedure • use formula $n\lambda = d\sin\theta$ • $n = 1$ since first order spectrum • find d using number of lines/mm = 300 mm⁻¹ • find θ using distance of grating from plastic ruler = 0.50 m and $x = 0.10$ m (not protractor) Determination of wavelength • calculate d ($= 10^{-3}/300$) = 3.3×10^{-6} m • use $x = 0.10$ m and distance to grating = 0.50 m to calculate $\tan \theta$ ($= 0.2$) • $\theta = 11.3^\circ$ • $\sin \theta = 0.196$ • alternatively, calculate hypotenuse of triangle (using Pythagoras's theorem) = 0.51 m, giving $\sin \theta$ ($= 0.10/2600^{1/2}$) = 0.196 • allow use of small angle rule ($\sin \theta \approx \tan \theta \approx \theta = 0.2$) • calculate λ ($= 0.196 \times 10^{-3}/300$) = 650 nm Estimation of uncertainty • negligible uncertainty in d (and n) • uncertainty in $\sin \theta$ is found using uncertainty in distance measurements • uncertainty in each distance measurement is ± 1.0 mm or ± 0.5 mm or ± 2.0 mm • maximum % uncertainty in $\tan \theta / \theta / \sin \theta = 3\%$ • so % uncertainty in $\lambda = \% \text{ uncertainty in } \sin \theta = 3\%$ <u>Examiner's Comments</u> Unfortunately, a significant number of candidates did not recognise the diffraction grating experiment here,

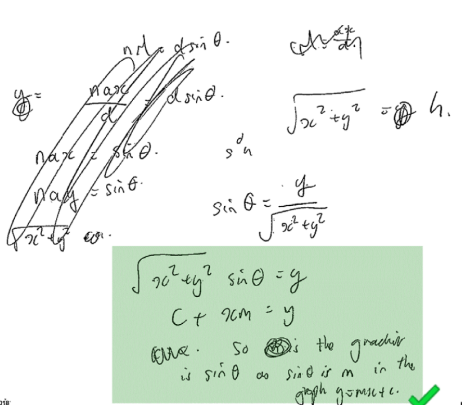
Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			confusing it with the double slit experiment and so using the formula $\lambda = ax/D$. This may be because the formula $n\lambda = d\sin\theta$ is in the astrophysics section of the formula sheet. Candidates who chose to use the correct formula $n\lambda = d\sin\theta$ were given for choosing the correct values for n, d and θ, for a correct calculation of λ, and for an accurate error analysis. Candidates who did not calculate λ could still gain full marks, as long as they gave accurate instructions as to how this could be done. Strong candidates successfully calculated a reasonable estimate of uncertainty in λ by combining the uncertainties in the distance measurements which had been used to find $\sin\theta$.  AfL The experiment to measure the wavelength of light using a diffraction grating is PAG 5.1 and so is often carried out in Year 12. It may be beneficial to carry out this practical activity in Year 13 instead during the study of spectral lines, to reinforce use of the formula $n\lambda = d\sin\theta$.  OCR support Being aware of the contents of the data, formulae and relationship booklet and its layout will support candidates, alleviating the need to recall numerical values of constants and allowing retrieval of correct formulae, or giving assurance that the student has recalled correctly. Exemplar 7

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					Red fringes occur only at the 4th order (a). There are 300 lines per millimetre, which is 300 000 per metre. Using slit separation is Need to measure the angle θ between the 3rd and 4th order to use equation $d \sin \theta = n \lambda$, rearrange for λ. A more fully decked room would be better as it's more easier to determine what the colours and see the lights. Exemplar 7 illustrates many aspects of a Level 1 response. Although the correct formula has been identified, it will not give a correct value for λ because incorrect values for n, d and θ have been chosen. The response has been put at the bottom of Level 1 because, although there is an attempt at a logical structure, almost all of the information it contains is inaccurate and therefore not relevant.
			Total	6	

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
6 a	$y = \sin(\theta) \sqrt{x^2 + y^2}$ compared with "y=mx+c"	B1	Allow: gradient = $\frac{\Delta y}{\Delta(\sqrt{x^2 + y^2})}$ with $\sin(\theta) = O/H$ Not: gradient = $\frac{y}{(\sqrt{x^2 + y^2})}$ unless "c=0" seen. Examiner's Comments Candidates found this item tricky even if they realised that $\sin(\theta) = y/\sqrt{x^2 + y^2}$ and then re-arranged the equation into a form comparable with the general equation of a straight line, "y=mx +c". Unless that comparison was clear, then the mark could not be credited. Exemplar 8  The exemplar shows a clear way of demonstrating how to show that the gradient of the line of the graph should be $\sin(\theta)$.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b	i	(Straight line of best fit showing) <u>gradient</u> = 0.73 $(d \sin \theta = n\lambda)$ $\frac{1.0 \times 10^{-3}}{600} \times 0.73 = 2 \times \lambda$ $\lambda = 6.1 \times 10^{-7} \text{ (m)}$	C1 C1 A1	Allow: gradient in range 0.70–0.76. Allow: evaluation of $\theta = 44\text{--}50$ (degrees) in place of gradient Allow: any subject Note: Gradient in range 0.70–0.76 gives λ in range $(5.8 - 6.4) \times 10^{-7} \text{ m}$ <u>Examiner's Comments</u> Many candidates could plot the best fit straight line and attempted to calculate the gradient. Not many candidates after that point realised that the gradient had given them $\sin(\theta)$ and could make no further meaningful progress. Common errors included not calculating d correctly from the quoted number of lines mm^{-1} or, less frequently, was using a value different from 2 for n.
		ii	(Scales/distances are large compared with the absolute uncertainty so) <u>absolute uncertainty is too small</u> to be shown (reasonably on this graph's scale) (AW)	B1	Ignore: error too small <u>Examiner's Comments</u> 20 per cent of candidates did not attempt this item. Some candidates were on the right lines but very few mentioned about absolute uncertainty and that for these instruments and this graph, the absolute uncertainty was too small to view on this scale.
		iii	(The values for λ or θ will be) less precise (as independent measurements less likely to agree) (AW)	B1	<u>Examiner's Comments</u> About two fifths of candidates appreciated that the precision would not be as good with a protractor, as repeated measurements would be less likely to cluster in close proximity. <u>Precision</u> The term 'precision' is defined of page 40 the Practical Skills Handbook, along with other useful terms that attempt to describe the quality of data
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
7	a	i	<u>Electron(s)</u> makes a transition to a lower (energy) level / loses energy and emitting a photon(s) / EM radiation	B1	Examiner's Comments Many candidates muddled up emission (lines emitted by a source) and omission (as in lines absorbed by a low pressure gas when a continuous spectrum passes through it, as in an absorption spectrum), so could not score the mark. Some focused on the experimental procedure of using a diffraction grating. A third of candidates correctly stated that the electron dropped down to a lower energy state, releasing a photon or the equally acceptable 'EM radiation'
		ii	Reduce grating separation / increase distance between grating and screen	B1	Allow 'use finer grating' or 'use grating with more lines mm^{-1}', Not 'smaller slit size' Examiner's Comments This item tested knowledge of specification 5.5.2 (g) and PAG 5. It would be advisable for Centres, where possible, to allow candidates to observe the effect of changing the slit separation and the grating-screen separation independently. Approximately a third of students correctly suggested making one of those adjustments, even if they expressed the former as 'increase the number of lines per mm'. Some candidates presented arguments about plotting the graph on a smaller scale or measuring the wavelength in picometres in an attempt to resolve the peaks in the plot, which was a mis-interpretation of the question.
		iii	wavelength (of peak) = 661.5 nm $v = 3.0 \times 10^8 \times (661.5 - 656.3) / 656.3$ recession velocity = $2.4 \times 10^6 \text{ (m s}^{-1}\text{)}$	C1 C1 A1	Allow: between 661 and 662 nm Note: check divided by 656.3 nm Range of acceptable answers. $2.1(5) - 2.6(1) \times 10^6$ Examiner's Comments Examiners were pleased to see this item answered well, with the majority of candidates gaining either two or three marks. Those that did not either misread the position of the red-shifted spectral peak, did not recognise that they were looking for the peak wavelength or did not use the 'at rest' wavelength for the denominator of the expression for the change in wavelength.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iv	(Relative) abundance of hydrogen (AW)	B1	Allow 'Hydrogen commonly found in stars' (AW) Examiner's Comments Just over half of all candidates realised that the useful property of hydrogen was its relative abundance in stars and hence galaxies.
	b		Less intense Galaxy is moving faster and therefore greater / longer wavelength (AW) Periodic shift in wavelength (if plane of orbit is in line of sight) (ORA)	B1 B1 B1	Allow 'greater red shift' / 'greater Doppler shift' / 'to the right' for longer wavelength Allow argument referring to splitting of line because of relative velocities of two component stars. Not idea of blue shift. Examiner's Comments Some 9% of all candidates declined to answer this item, the highest rate for any item on this paper. The most common correct response linked higher distance with higher recessional velocity and thus higher increase in wavelength. Higher ability candidates explained that the orbiting stars would have different velocities relative to the Earth resulting in a periodic change in wavelength from the central peak. References to blue-shifting were erroneous and contradictory.
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8		i	The (total radiant) power (of a star) /AW	B1	
		ii	$(L = 4\pi r^2 \sigma T^4)$ $\text{ratio} = \sqrt{\frac{6.92 \times 7500^4}{10.0 \times 4500^4}}$ ratio = 2.3(1)	C1 A1	Allow 1 mark for 5.3; square root omitted Allow 1 mark for 1:2.3 or 0.43
		iii	$\lambda_{(\max)} \propto \frac{1}{T}$ Lower temperature star will have the longest wavelength, so it is Aa2	B1 B1	Allow word equation Note Must mention wavelength <u>and</u> temperature
		iv	From their (different) colours	B1	
		v	Any <u>three</u> from: • Continuous spectrum • Light / radiation / photons passes through cooler gas/star's atmosphere • Photon(s) absorbed by electron(s) • Electron(s) excited / jump / make transition to higher energy level(s) • Electron only promoted if energy of photon matches energy gap between two given levels • Photons remitted in different directions • (so) idea of contrast with non-absorbed wavelengths	B1 × 3	
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
9		i	Dark lines are absorption lines / idea of photon absorption / idea of electrons absorbing energy / idea of electrons making transition to higher energy level (i.e. excitation) Caused by light passing through (cooler) stellar atmosphere or Earth's atmosphere	B1 B1	Allow (cooler) gases/hydrogen (gas) <u>Examiner's Comments</u> Apart from a small number of candidates that confused emission spectra with absorption spectra, it was clear that most candidates recognised that what they needed to do. Technical language skills prevented some otherwise sensible answers from scoring 2 marks.
		ii	$\Delta \lambda/\lambda = 22/410$ or $21/411$ $v = 0.054c = 1.6(1) \times 10^7 \text{ (m s}^{-1}\text{)}$	C1 A1	NB $21/411$ gives $1.5 \times 10^7 \text{ (m s}^{-1}\text{)}$ Any evidence of use of 432 as denominator of $\Delta \lambda/\lambda$ is XP zero marks <u>Examiner's Comments</u> Most candidates selected the correct formula in this part. If a candidate responded at all, they were likely to score both marks. A small percentage of candidates used the wrong wavelength for the denominator of the relevant expression. This was marked as 'wrong Physics' (XP) on scripts.
		iii	Big Bang / Expanding universe i.e. Hubble's Law / dark matter or dark energy / AW	B1	ignore redshift ignore Doppler <u>Examiner's Comments</u> An overwhelming majority of candidates made some reference to either the Big Bang Theory, cosmological expansion or simply 'Hubble's Law'. Some other answers were also given credit as shown in the Mark Scheme although these were very much rarer.
			Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
10		i	$E = hc/\lambda$; $\Delta\varepsilon = E_1 - E_2 = hc\Delta\lambda/\lambda^2$	C1	allow calculation of $E = hc/\lambda$ twice and difference taken
		i	$\Delta\varepsilon = 6.63 \times 10^{-34} \times 3 \times 10^8 \times 0.6 \times 10^{-9} / 5.9^2 \times 10^{-14}$	C1	
		i	$\Delta\varepsilon = 3.4 \times 10^{-22} \text{ (J)}$	A1	
		ii	$\sin \theta = n\lambda/d$; $1/d = 3 \times 10^5 \text{ (m}^{-1}\text{)}$	C1	or similar allow 20.72 – 20.70
		ii	$\theta_1 - \theta_2 = \sin^{-1} (2 \times 589.6 \times 3 \times 10^{-4}) - \sin^{-1} (2 \times 589 \times 3 \times 10^{-4})$	M1	
		ii	$\theta_1 - \theta_2 = 20.717 - 20.695 = 0.022^\circ$	A1	
			Total	6	
11		i	$\Delta\lambda = \frac{\lambda v}{c} = \frac{486 \times 10^{-9} \times 960 \times 10^3}{3.00 \times 10^8}$	C1	
		i	$\Delta\lambda = 1.56 \text{ (nm)}$	C1	
		i	$\lambda = 486 + 1.56 = 488 \text{ (nm)}$	A1	
		ii	$d = 1.25 \times 10^{-6} \text{ m}$ $\theta = \sin^{-1} \left(\frac{2 \times 486 \times 10^{-9}}{1.25 \times 10^{-6}} \right)$	C1	Allow 1 mark $\theta = \sin^{-1} \left(\frac{2 \times 488 \times 10^{-9}}{1.25 \times 10^{-6}} \right) = 51^\circ$; incorrect 488 nm used instead of 486 nm.
		ii	$\theta = 51^\circ$	A1	
			Total	5	
12			D	1	
			Total	1	